

WHO'S WHO

The selection of people included in this **Who's Who** reflects the main experimenters, philosophers, and scientists represented in the book. Cross references have been included for scientists with biography panels within the main timeline pages.

Alhazen (965–1040) Arab mathematician, astronomer, and physicist widely considered to be the father of modern optics. Arguing that light enters the eye rather than being emitted from it, his influential treatise *Optics* described the laws of reflection and refraction, as well as the anatomy of the human eye. He also tried to develop realistic cosmological models.

Al-Khwarizmi (c.780–c.850) Persian mathematician, geographer, and astronomer responsible for introducing Hindu–Arabic numerals and algebra to the West. Working in Baghdad's translation and research center, the House of Wisdom, he produced two mathematical textbooks and updated Ptolemy's *Geography*, presenting coordinates for places around the world. The word "algorithm" is derived from the Latin pronunciation of Al-Khwarizmi's name.

Al-Kindi (Abu Yusuf Ya'qub Ibn 'Ishaqal-Kindi), Arab philosopher (c.801–c.873) See p.46

Al-Razi (Rhazes), Arab philosopher (c.865–c.925) See p.48

Alvarez, Luis Walter, American physicist (1911–88) See p.311

Ampère, André-Marie, French mathematician and physicist (1775–1836) See p.181

Ångström, Anders Jonas (1814–74) Swedish physicist and father of spectroscopy who discovered that hot gas emits and absorbs light at the same wavelengths at which it absorbs light when cooler. Ångström wrote on heat, magnetism, optics, and the solar spectrum and was the first to examine the spectrum of the aurora borealis. The Ångström unit (Å) for measuring atomic distances is 10^{-10} m.

Anning, Mary, British fossil hunter (1799–1847) See p.176

Archimedes (c.290–c.212 BCE) Greek inventor, philosopher, and mathematician, who stated that any object immersed in a liquid will experience an upward force equal to the weight of the liquid displaced. Archimedes wrote works on arithmetic, geometry, and mechanics, and constructed siege machines to defend Syracuse against the Romans. He is also credited with the creation of the Archimedes screw water pump.

Aristarchus of Samos (c.310–230 BCE) Greek astronomer who first suggested that Earth revolves around the Sun. Aristarchus's treatise *On the Sizes and Distances of the Sun and Moon* incorrectly calculated the Sun as 20 times as far from Earth as the Moon, and 20 times the size of the Moon, but his pioneering methods paved the way for future astronomical studies.

Aristotle, Greek philosopher (384–322 BCE) See p.29

Arkwright, Richard (1732–92) English textile industrialist whose water frame invention

enabled the automated spinning of cotton threads. Arkwright installed his water frames in specially built factories, an early example of mass production and the Industrial Revolution.

Arrhenius, Svante August (1859–1927) Swedish physicist and chemist who was awarded the Nobel Prize in Chemistry for his electrolytic theory of dissociation. He was also the first to recognize that carbon dioxide in the atmosphere could create a greenhouse effect on Earth's surface. The lunar crater Arrhenius is named after him.

Avicenna (see Ibn Sina)

Avogadro, Amedeo (1776–1856) Italian mathematical physicist whose law states that equal volumes of gases contain equal numbers of molecules when at the same temperature and pressure. As a tribute, the number of elementary particles in a mole was called Avogadro's constant.

Babbage, Charles (1791–1871) English mathematician and inventor regarded in Britain as the pioneer of modern computers. Babbage devoted his life to building two mechanical calculating machines, including his Analytical Engine, designed to perform arithmetic using punched cards as its memory source. Neither machine was successfully completed.

Bacon, Francis, English philosopher (1561–1626) See p.98

Bacon, Roger, English scholar (c.1220–92) See p.60

Baekeland, Leo Hendrik (1863–1944) Belgium-born American chemist who invented Velox, the first photographic paper that could be developed under artificial light. In 1899, Baekeland sold his Velox rights to American innovator George Eastman for \$1 million and used the proceeds to develop his most famous invention, Bakelite—the first synthetic plastic that could be poured into molds to harden in different shapes.

Baird, John Logie (1888–1946) Scottish engineer, inventor, and television pioneer. Baird first televised objects in 1924, moving objects in 1926, and produced the first color transmission in 1928. When the BBC began broadcasting in 1936, Baird's mechanical scanning system competed with Guglielmo Marconi's EMI electronic system, which the corporation adopted exclusively from 1937.

Banks, Joseph (1743–1820) English botanist, naturalist, president of the Royal Society, and often called Australia's first scientist. Banks traveled around the world on Captain Cook's HMS *Endeavour* and introduced many new plants to the West. Geographical features and plants have been named after him. He helped establish the Royal Botanic Gardens at Kew, London, and persuaded the government to invest in scientific exploration.

Bardeen, John (1908–91) American physicist awarded the Nobel Prize in Physics twice: in 1956 for co-inventing the transistor; and in 1972 for developing the theory of superconductivity. The transistor paved the way for modern electronics, while superconductivity was used

in medical advances such as MRI (magnetic resonance imaging). Bardeen was professor of electrical engineering and physics at Illinois University from 1951 to 1975.

Barnard, Christiaan Neethling (1922–2001) South African surgeon who performed the first human heart transplant. Barnard introduced open heart surgery, performing heart transplants on dogs and designing an artificial heart valve. In 1967 he performed the world's first human heart transplant on grocer Louis Washkansky, who later died from pneumonia.

Bassi, Laura, Italian physicist (1711–78) See p.137

Becquerel, Antoine-Henri (1852–1908) French physicist who shared the 1903 Nobel Prize in Physics with Marie and Pierre Curie for discovering radioactivity. He discovered radioactivity accidentally by experimenting with phosphorescence and uranium salts. This led to the isolation of radium and paved the way for modern nuclear physics.

Bell, Alexander Graham, American inventor (1847–1922) See p.217

Bell Burnell, Jocelyn, British astrophysicist (1943–) See p.296

Benz, Karl (1844–1929) German inventor who, together with Gottlieb Daimler, created the first gasoline-powered motor vehicle. Benz patented his three-wheeled, four-stroke cylinder Motorwagen in 1886 and produced the first four-wheel automobile in 1893. This laid the foundation for the motor industry and in 1899 Benz & Co. began producing the world's first racing cars.

Berg, Paul (1926–) American biochemist and co-winner of the 1980 Nobel Prize in Chemistry for developing recombinant DNA techniques for splicing and recombining DNA from different organisms, which led to modern genetic engineering.

Berners-Lee, Tim, British computer scientist (1955–) See p.324

Bernoulli, Daniel (1700–82) Swiss physicist and mathematician, who proposed that the pressure in a fluid decreases as the speed of its flow increases—Bernoulli's principle. Bernoulli's 1738 work *Hydrodynamica* was very important for kinetic theory of gases and fluids, and proposed practical applications of watermills, water propellers, and water pumps. Bernoulli also investigated medicine, biology, astronomy, and oceanography.

Bernoulli, Johann, Swiss mathematician (1667–1748) See p.124

Berzelius, Jöns Jakob (1779–1848) Swedish chemist considered to be the founding father of modern chemistry. Berzelius is noted for formulating his electrochemical theory, producing a list of atomic weights, and developing modern chemical symbols. A professor of medicine and member of the Royal Swedish Academy of Sciences, Berzelius discovered and isolated several elements, developed analytical techniques, and investigated isomerism and catalysis.

Bessemer, Henry (1813–98) English engineer who introduced the Bessemer process for creating the first inexpensive steel by blowing air through molten iron. The son of a metallurgist, Bessemer manufactured gold paint powder, invented a sugarcane crushing machine, and developed a cast-iron cannon for the Crimean War.

Biot, Jean-Baptiste (1774–1862) French physicist who established the existence of meteorites and made the first scientific balloon flight for scientific purposes. He won a Royal Society award for his study of light polarization and helped develop saccharimetry, a technique for analyzing sugar solutions. Working with fellow physicist Félix Savart, he formulated the Biot–Savart law, a fundamental component of modern electromagnetic theory.

Bjerknes, Vilhelm (1862–1951) Norwegian meteorologist and physicist who helped found modern weather forecasting. As a professor at Stockholm University, Sweden, Bjerknes studied hydrodynamics and thermodynamics and their relation to atmospheric motion. This led to the theory of air masses, an essential component of modern-day weather forecasting. He later founded the Geophysical Institute and Weather Service of Bergen in Norway.

Black, Joseph (1728–99) Scottish chemist and physician famous for discovering that fixed air (carbon dioxide) is present as a distinct gas in the atmosphere. He also discovered latent heat by showing that when ice melts, it takes up heat without changing temperature.

Bode, Johann Elert (1747–1826) German astronomer responsible for Bode's law (Titus–Bode rule) predicting the relative spacing between the Sun and its planets.

Bohr, Niels (1885–1962) Danish physicist awarded the 1922 Nobel Prize in Physics for using quantum theory to explain atomic structure. Bohr's 1913 model of the atom describes a central atomic nucleus with electrons in orbit around it. Bohr joined the Manhattan Project during World War II, but later advocated the peaceful use of nuclear energy.

Bonnet, Charles, Swiss naturalist and philosopher (1720–93) See p.148

Boole, George (1815–64) English mathematician who pioneered Boolean algebra—symbolic logic and the rules that govern it. His ideas proved vital for modern computer science.

Bosch, Carl (1874–1940) German chemist whose Haber–Bosch process for the high pressure synthesis of ammonia won him the 1931 Nobel Prize in Chemistry, and is today the standard industrial procedure for nitrogen fixation.

Bose, Satyendranath (1894–1974) Indian mathematician and physicist who collaborated with Albert Einstein in the study of quantum mechanics. Together, they developed Bose–Einstein statistics for studying the behavior of bosons (particles with integral spin values named after Bose), important for lasers and superfluid helium.